ZYK-558 · DARPA D2 SPRINT ENTRY PACKAGE DRAFT · OCTOBER 24, 2026

AUTONOMOUS DOCUMENTATION UNDER CHAOS

Entry package draft for the DARPA Documentation and Decision Support (D2) Sprint. $1,000,000 prize pool. Fully virtual: teams download datasets, build on their own systems, submit algorithms. Final submissions due March 1, 2027. Team of two, which the rules permit. Her hand submits.

1. THE PROBLEM (DARPA'S FRAME, HELD EXACTLY)

Roughly 75% of combat casualty care goes unrecorded. Not badly recorded - unrecorded: the medic's hands are occupied, the environment is loud, degraded, and lethal, and documentation loses to hemorrhage control every time. The consequence compounds: the next provider re-derives history from wounds, tourniquet times are lost, and the decision-support systems the military builds have no ground truth to learn from. The D2 Sprint asks for AI that autonomously documents care and supports TCCC (Tactical Combat Casualty Care) decisions in real time.

2. OUR THESIS

Documentation is a DETERMINISM problem, not a generation problem. A system that paraphrases what it thinks it saw will fabricate under uncertainty, and in casualty care a fabricated vital sign is a casualty. Our system never generates an ungrounded clinical fact: every recorded element is bound to a sensor observation with a timestamp and a confidence bound, and anything below the bound is recorded AS UNKNOWN - explicitly, structurally, in the record itself. The documentation layer speaks the TCCC vocabulary natively (MARCH sequence, timing, interventions) so the record is decision-ready the moment it is written.

3. ARCHITECTURE

Three layers. (1) CAPTURE: passive multi-sensor fusion - audio, video, physiologic waveforms, device telemetry - running at the edge, offline-first, because connectivity is a casualty of the same environment. (2) STRUCTURE: a deterministic event-extraction layer, built on our formal-language core (DELTA, five levels shipped, 82/82 tests), converts observations into timestamped clinical events with provenance on every field. (3) SUPPORT: the TCCC decision-support layer presents protocol-aligned options with the evidence trail attached; it recommends, the human decides, and the record carries both. The verification layer re-derives every summary from the raw event stream before release - the machine can be audited against its own sensors.

4. EXECUTION PLAN (WE HAVE DONE THIS SHAPE BEFORE)

Nov-Dec 2026: datasets downloaded, baseline pipeline on the provided evaluation data. Jan 2027: event-extraction layer hardened against the sprint's scoring rubric; refusal behavior tuned. Feb 2027: full-dataset validation runs, ablations, submission package. Final submission before March 1, 2027. The team: principal architect (information architecture, clinical logic, the method) + second principal (tooling, verification, the record). Two-person teams are lawful under the sprint rules; the lab's whole method is built for exactly this headcount.

5. WHY THIS TEAM WINS THE ROOM

The lab's founding competency is documentation under chaos: a 68-batch evidentiary archive processed into a 269-page compendium, 1,512 commits, 602 briefings - with hash-verified integrity and a fixed error-classification system. We build records that survive. That is the entire ask of the D2 Sprint, pointed at the battlefield instead of the archive.

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